

Skills

Programming Languages: C++, Java, Python, TypeScript, JavaScript (ES6), SQL

Web Technologies: HTML, CSS, React, REST API, GraphQL API, Microservices, Node.js, Express.js, Sequelize

Database: RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse

Cloud: AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs)

Tools: Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Kubernetes, Power BI, QuickSight, Spring Boot

AI/ML: LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL.

Experience

Campus Mesh

Jan 2026 – Present

Software Engineer

New York, US

- Developed and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing user engagement by **28%** and retention by **50%**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
- Implemented a **Model Context Protocol (MCP)** server for efficient tool-augmented retrieval, cutting AWS Athena query time and compute costs by **40%**.
- Built a high-throughput, event-driven incident triage system on **AWS ECS Fargate, SQS, SNS**, decoupling distributed message processing and reducing response latency by **50%**.
- Provisioned resilient **Infrastructure-as-Code** using **AWS CDK** on auto-scaling ECS clusters, ensuring **99.9%** availability under 2x peak load.

New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

Software Developer (On-Campus)

New York, US

- Architected an **NSF-funded (\$100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrics analysis and food image processing.
- Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
- Developed an insurance analytics platform (**React, Node.js, REST APIs**); deployed **Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **\$50K+** revenue.
- Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

- Developed a distributed backend on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing and fault tolerance.
- Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.
- Deployed an **NLP based OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.
- Optimized data ingestion with **parallel processing** for **100K-row** from Excel; reduced execution time from **5 hours to 1 hour**.
- Built **Power BI dashboards** processing large datasets from multiple databases, improving reporting efficiency by **40%**.

Projects

Advanced Expense Tracker (Java Desktop Application) 🌐 | *Java, JavaFX, JDBC, SQLite, JFreeChart*

- Architected a modular **Java application** supporting individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns**.
- Implemented concurrent transaction handling and real-time cost splitting using **multithreading** and synchronized blocks to ensure thread-safe database updates.

Hospital Management System (Full-Stack Application) | *React, Node.js, Express, MongoDB, SageMaker, Hugging Face*

- Built a **full-stack healthcare platform** using **React, Node.js, Express, and MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
- Fine-tuned a **Hugging Face transformer-based model** achieving **88% accuracy** for health condition prediction and deployed it on **AWS SageMaker** for real-time diagnostic support in the doctor dashboard.

Education

New York University | *Master of Science in Computer Engineering (GPA – 3.9/4)*

Sep 2024 – May 2026

Certifications 🌐

AWS Cloud Practitioner 🌐

Power BI Data Analyst Associate 🌐

Azure Data Fundamentals 🌐

Microsoft Developer Associate 🌐

Microsoft Solution Architect Expert 🌐

Stanford Machine Learning (Coursera) 🌐